Digital Discovery | *Paper*

PhaseFoundry: An Open-Source CALPHAD-Coupled Phase-Field Framework for Spinodal Microstructure Design in Multicomponent Alloys

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Abstract. Predicting how a multicomponent alloy decomposes into a nanoscale two-phase microstructure requires connecting thermodynamic databases, which encode where a solid solution is unstable, to phase-field simulations, which resolve how that instability evolves in space and time. In practice this connection is the weakest link in the integrated computational materials engineering (ICME) toolchain: free energy models are hand-transcribed into solver source code, chemical potentials are approximated by finite differences, and coherency strain effects are either neglected or parameterized *ad hoc*. We present `PhaseFoundry`, an open-source Python framework that closes this gap for spinodal microstructure design. Starting from a CALPHAD thermodynamic database—including databases generated from machine-learned interatomic potentials (MLIPs) via `PhaseForge`—the framework evaluates exact Gibbs energy Hessians and chemical potentials using analytic Jansson derivatives in `pycalphad`, maps spinodal instability regions through gauge-invariant eigenanalysis, and evolves the resulting instabilities with an unconditionally stable pseudo-spectral Cahn–Hilliard solver that optionally includes coherency strain via a Moulinec–Suquet spectral micromechanics solver with MLIP-derived, composition-dependent elastic constants. Binary, ternary, and quaternary alloy systems are specified entirely through YAML configuration files, with no code changes. `PhaseFoundry` is available under the MIT license at `https://github.com/vahid2364/PhaseFoundry`.

1 Introduction

Spinodal decomposition, the barrierless unmixing of a supersaturated solid solution into compositionally modulated domains by uphill diffusion, has been understood thermodynamically since Gibbs[1] and given its modern continuum formulation by Cahn and Hilliard.[2,3] It has recently reemerged as a deliberate microstructure design strategy. Depending on the alloy, the transformation can be harmful, embrittling otherwise ductile solid solutions[4] or providing a pathway for deleterious ordered phases,[5] or it can be exploited: in refractory and 3d-transition-metal high-entropy alloys (HEAs), spinodally modulated solid solutions have been shown to overcome the strength-ductility trade-off,[6–8] tailor magnetic properties,[9,10] and improve thermoelectric performance.[11] Either way, the designer must first locate the spinodal region within a vast, high-dimensional composition space, and then predict the morphology, length scale, and phase compositions of the resulting microstructure.

The thermodynamic half of the problem is well served by mature tools. CALPHAD databases encode assessed Gibbs energy models for multicomponent phases, and both commercial engines such as Pandat[12] and open-source engines such as `pycalphad`[13] make these models programmatically accessible, with ESPEI providing automated database development and uncertainty quantification.[14] Where classical assessments do not exist, MLIP-based workflows such as `PhaseForge`[15–17] can now generate CALPHAD-compatible TDB files for arbitrary alloy systems at a fraction of the cost of density functional theory.[18] The stability analysis needed to locate spinodal regions has a long lineage, from de Fontaine's coherent stability criteria[19,20] through Gibbs energy Hessian analysis for HEAs[21] and high-throughput CALPHAD-based spinodal screening.[22,23] The kinetic half is likewise well served by general-purpose phase-field frameworks such as MOOSE,[24] PRISMS-PF,[25] and FiPy.[26]

The difficulty lies in the coupling. A quantitative spinodal simulation needs the derivatives of the CALPHAD molar Gibbs energy evaluated millions of times per run. General-purpose frameworks require users to supply free energy functions explicitly, which for multicomponent phases means either hand-fitting polynomial surrogates that drift from the database or numerically differentiating equilibrium calculations—an approach sensitive to stencil size in high-dimensional composition spaces.[27] Coherency strain, which

shrinks the effective spinodal region[3,20] and selects decomposition morphology, adds a further layer of parameterization. To our knowledge, no open-source tool provides an end-to-end pipeline from a CALPHAD TDB to a running elasto-chemical phase-field simulation for multicomponent alloys.

`PhaseFoundry` provides that pipeline. Its central design decisions are:

1. **Analytic thermodynamic derivatives.** Chemical potentials and Gibbs energy Hessians are computed with the Jansson derivative method,[27] eliminating finite-difference error and stencil-size sensitivity.
2. **Gauge-invariant stability analysis.** Spinodal regions are identified from eigenvalues of the Hessian in a Cartesian basis,[21,23] visualized with affine projections.[28]
3. **Unconditionally stable spectral solver.** The Cahn–Hilliard equations are advanced with semi-implicit Eyre convex splitting[29] with per-component stabilization constants.
4. **MLIP-informed elasticity.** Coherency strain enters through a Vegard eigenstrain model solved with the Moulinec–Suquet spectral method,[30] with elastic constants from SQS/MLIP calculations.[31]
5. **Configuration over code.** A single YAML file specifies the entire run; binary, ternary, and quaternary systems in 2D or 3D share one code path.

A companion paper[32] presents the scientific application to the Hf–Nb–Ti–V refractory HEA system.

2 Framework overview

2.1 Pipeline

`PhaseFoundry` is organized as an installable Python package, `calphad_pf`, driven by numbered workflow scripts sharing a single YAML configuration file per run (Fig. 1):

- **Step 1** samples the CALPHAD TDB over a composition grid, evaluating G_m^α, relative chemical potentials, and Jansson derivatives with `pycalphad`, storing results as HDF5 tables.
- **Step 1b** (elasto-chemical) tabulates SQS/MLIP elastic constants C_{ij} and lattice parameters from `PhaseForge` CSV outputs.
- **Step 1c** (elasto-chemical) computes CALPHAD tie-line endpoint compositions at the simulation temperature.
- **Step 2** fits a scattered-data interpolant surrogate and computes spinodal boundaries from gauge-invariant Hessian eigenvalues.
- **Step 3** runs the pseudo-spectral Cahn–Hilliard simulation with optional elasto-chemical coupling and GPU execution.
- **Step 4** post-processes snapshots and extracts composition, stress, and strain fields.
- **Steps 5–6** are verification stages: Step 5 validates derivative methods against direct `pycalphad` evaluations; Step 6 compares chemical-only and elasto-chemical runs.

The full pipeline executes with a single command:

```
python workflows/run_all.py \
  --config configs/nb_hf_ti_v_elastic.yaml
```

2.2 Package architecture

Three subpackages separate concerns cleanly. `calphad_pf.thermodynamics` handles TDB sampling, HDF5 storage, Hessian eigenanalysis, spinodal boundary extraction, and the interpolant surrogate. `calphad_pf.phasefield` contains the Cahn–Hilliard solver, the Eyre-stabilized spectral update, FFT operations (NumPy/CuPy backend), and visualization utilities. `calphad_pf.elasticity` implements the eigenstrain models, the SQS elastic-constant interpolant, and the Lippmann–Schwinger spectral solver. The dependency stack is deliberately shallow: NumPy, SciPy, h5py, scikit-learn, Matplotlib, and `pycalphad`, with CuPy as an optional GPU extra.

Because the solver consumes only the surrogate interface, thermodynamic sampling and simulation are decoupled. A new alloy system requires only a new TDB and a new YAML file.

3 Numerical methods and implementation

3.1 CALPHAD thermodynamics with analytic derivatives

For an n-component phase α with the n-th component as the dependent variable, local stability is governed by the Hessian $H_{ij} = \partial^2 G_m^\alpha / \partial x_i \partial x_j$: the phase is spinodally unstable wherever at least one eigenvalue of H is negative.[3,21] `PhaseFoundry` computes these derivatives analytically with the Jansson method[27] as implemented in `pycalphad`. The `finite_diff` and `hybrid` modes are selectable via the configuration file for validation and backward compatibility.

The Hessian is transformed to a Cartesian basis before eigenanalysis to yield gauge-invariant eigenvalues independent of the choice of dependent component.[23] For quaternary and higher systems, instability maps are displayed with affine projections.[28] Step 2 wraps the HDF5 tables in a scattered-data interpolant surrogate—the only thermodynamic object accessed at run time—fit once per system and temperature.

3.2 Phase-field solver

The solver evolves $n-1$ independent composition fields $x_i(\mathbf{r},t)$ under the free energy functional

$$\mathcal{F}[\mathbf{x}] = \int_\Omega \left[G_m^\alpha(\mathbf{x}) + \sum_{i=1}^{n-1} \frac{\kappa_i}{2} |\nabla x_i|^2 \right] dV, \tag{1}$$

via the coupled Cahn–Hilliard equations

$$\frac{\partial x_i}{\partial t} = \nabla \cdot \sum_j M_{ij}(\mathbf{x})\, \nabla \mu_j^{\mathrm{tot}}, \tag{2}$$

with $\mu_j^{\mathrm{tot}} = \partial G_m^\alpha / \partial x_j - \kappa_j \nabla^2 x_j + \mu_j^{\mathrm{el}}$. The mobility is degenerate, $M_{ij} = \mathring{M}_{ij} \prod_k x_k$, and the bare matrix $\mathring{M}_{ij}$ is user-supplied, accommodating strongly asymmetric diffusivities across components.

Following Eyre,[29] the molar Gibbs energy is split into convex and concave parts, with the convex quadratic part $\frac{A_i}{2} x_i^2$ treated implicitly. The Fourier-space update is

$$\hat{x}_i^{n+1} = \frac{\hat{x}_i^n + \Delta t\, k^2 \sum_j M_{ij}^{\mathrm{eff}} \hat{\mu}_{j,\mathrm{bulk}}^n - \Delta t\, k^2 M_{ii}^{\mathrm{eff}} A_i \hat{x}_i^n - \Delta t\, k^4 \sum_j M_{ij}^{\mathrm{eff}} \kappa_j \hat{x}_j^n}{1 + \Delta t\, k^2 M_{ii}^{\mathrm{eff}} A_i + \Delta t\, k^4 M_{ii}^{\mathrm{eff}} \kappa_i}, \tag{3}$$

where $A_i \geq \max(0, -H_{ii}(\mathbf{x}_0))$ is the per-component stabilization constant. The denominator is strictly positive for any $\Delta t > 0$, making the scheme unconditionally stable. Setting `dt: null` activates automatic time-step estimation from the largest Hessian eigenvalue.

The same code path runs in 2D and 3D. Setting `use_gpu: true` moves all FFT and array operations to a CUDA device via CuPy. Long runs write incremental NPZ checkpoints.

3.3 Elasto-chemical coupling

When enabled, mechanical equilibrium is solved on the current composition field with the Moulinec–Suquet spectral scheme,[30] iterating a Lippmann–Schwinger equation in Fourier space. The resulting stress field defines the elastic chemical potential μ_k^{el} following Gururajan and Abinandanan.[33]

Two eigenstrain models are implemented. The `fixed_vegard` model uses Vegard coefficients and stiffness tensors fixed at the CALPHAD tie-line endpoints from Step 1c. The `local_vegard` model evaluates the eigenstrain pointwise:

$$\varepsilon^*(\mathbf{r}) = \frac{a(\mathbf{x}(\mathbf{r})) - a_0}{a_0}, \tag{4}$$

with $a(\mathbf{x})$ and $\mathbf{C}(\mathbf{x}(\mathbf{r}))$ interpolated from SQS/MLIP tables across the full composition space. This model has the physically correct property that the eigenstrain vanishes in the initial homogeneous alloy. Elastic constants from ORB-v2 and GRACE-2L are shipped as data tables; the CSV interface accommodates other potentials.

The elastic solve runs every n_{el} time steps with warm starting, and the elastic potential is cached between solves.

3.4 Verification and validation tooling

Step 5 regenerates thermodynamic quantities directly from `pycalphad` at randomly sampled compositions and compares them to the surrogate, quantifying interpolation error and the discrepancy between derivative modes. Fig. 2 shows this comparison for the Nb–V system at $500\,^\circ\mathrm{C}$, where all three derivative modes agree closely across the full composition range. Step 6 runs matched chemical-only and elasto-chemical simulations from identical initial conditions, isolating coherency strain effects on the decomposition path. A `pytest` suite covers thermodynamic sampling, surrogate fitting, spectral operators, and the elasticity solver.

4 Illustrative applications

Binary Nb–V. The Nb–V BCC_A2 system serves as the reference case. Chemical-only simulations at $500\,^\circ\mathrm{C}$ across mean compositions $x_{\mathrm{Nb}} = 0.3$–0.8 show the expected progression from isolated to bicontinuous morphologies, in both 2D (128^2) and 3D (128^3) domains. Fig. 3 shows a representative equiatomic microstructure. Adding elastic coupling refines and aligns the modulations, consistent with classical coherent spinodal theory.

Ternary Nb–Hf–V. With two independent composition fields, anisotropic mobilities (Hf roughly two orders of magnitude slower than Nb), and distinct gradient energy coefficients per field, the ternary configurations demonstrate multicomponent capabilities: the two fields decompose on different time scales, and the gauge-invariant stability analysis maps composition-dependent instability directions across the Gibbs triangle.

Quaternary Hf–Nb–Ti–V. The equiatomic elasto-chemical configuration runs 3×10^6 time steps with three coupled fields, MLIP-derived composition-dependent stiffness, and tie-line endpoints informed by the experimental XRD phase compositions of An *et al.*[6] Scientific analysis of the results appears in the companion paper.[32]

Across all cases the practical point is uniformity: the same package, solver, and post-processing scripts handle every system; only the YAML file changes.

5 Relation to existing software

`PhaseFoundry` is complementary to, not a replacement for, general-purpose phase-field frameworks. MOOSE[24] and PRISMS-PF[25] offer massively parallel finite element infrastructure and broad multiphysics coverage, but leave multicomponent free energy construction and elastic parameterization to the user. FiPy[26] provides flexible PDE specification at a lower performance ceiling. Commercial couplings between CALPHAD engines and phase-field codes exist (e.g., MICRESS with Thermo-Calc) but are closed-source and not extensible to MLIP-generated databases. `PhaseFoundry` occupies the specific niche of open, reproducible, TDB-to-microstructure spinodal simulation, trading geometric and

multiphysics generality for a fully automated and thermodynamically exact pipeline.

6 Limitations and future directions

The current implementation assumes a single solid solution phase on periodic uniform grids under isothermal conditions. Nucleation of distinct crystal structures, polycrystalline domains, and non-conserved order parameters lie outside the model. The elastic solver neglects the stiffness-contrast energy correction and `local_vegard` evaluates Vegard coefficients at the parent composition as a first-order approximation. Mobility matrices are user-supplied constants; coupling to CALPHAD kinetic databases is a natural next step. Planned developments include temperature-stepped annealing schedules, DICTRA-style mobility database ingestion, and extension of stability analysis to quinary systems.

7 Conclusions

`PhaseFoundry` automates the path from a CALPHAD thermodynamic database to quantitative spinodal microstructure predictions in multicomponent alloys, combining analytic Jansson derivatives, gauge-invariant stability analysis, an unconditionally stable spectral Cahn–Hilliard solver, and MLIP-informed elasto-chemical coupling behind a single YAML configuration interface. Together with `PhaseForge`, it completes an open-source chain from machine-learned interatomic potentials to microstructure for HEA design.

Data and code availability

`PhaseFoundry` is available under the MIT license at `https://github.com/vahid2364/PhaseFoundry`, including all YAML configurations, workflow scripts, SQS/MLIP elastic constant tables, and post-processing scripts. Thermodynamic databases were generated with `PhaseForge`[15] (DOI: 10.5281/zenodo.15730911).

Author contributions

V.A.: conceptualization, software, methodology, validation, visualization, writing (original draft). R.A.: conceptualization, funding acquisition, supervision, writing (review and editing).

Conflicts of interest

There are no conflicts of interest to declare.

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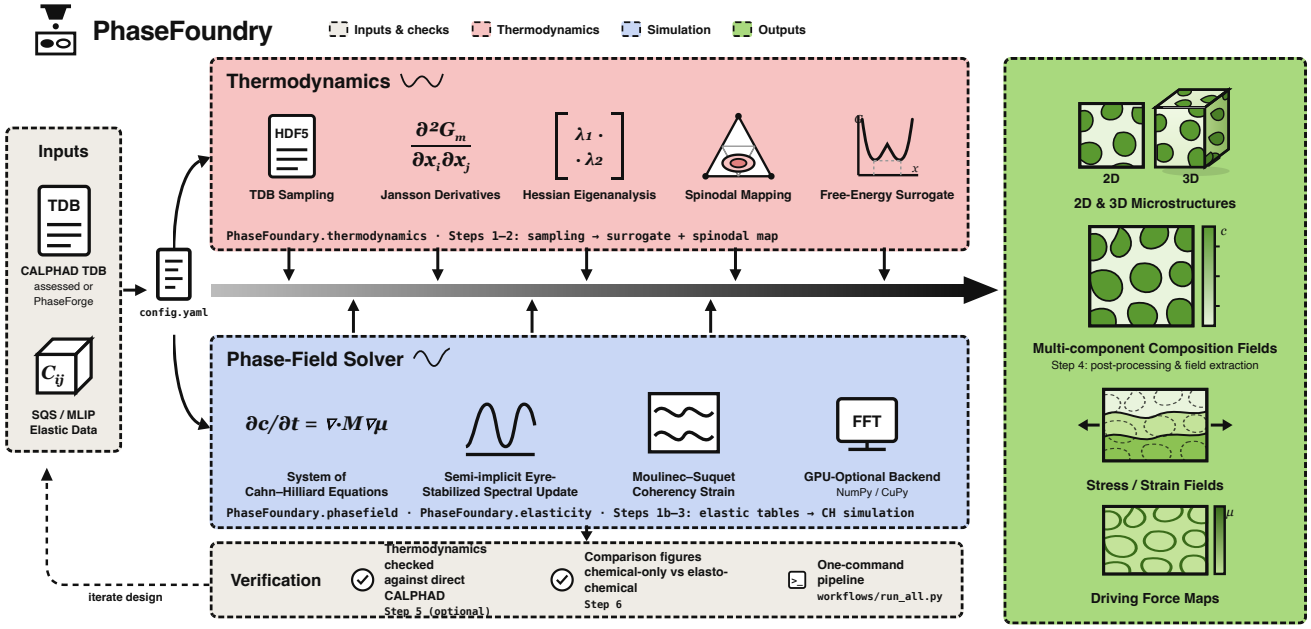


Fig. 1 | Workflow of *PhaseFoundry*. From a CALPHAD TDB (assessed or MLIP-generated via PhaseForge), SQS/MLIP elastic data, and a single YAML configuration, PhaseFoundry samples the database with analytic Jansson derivatives, maps spinodal regions through gauge-invariant Hessian eigenanalysis, fits a free-energy surrogate, and evolves the instability with an Eyre-stabilized pseudo-spectral Cahn–Hilliard solver with optional Moulinec–Suquet coherency strain, producing spinodal microstructures, composition fields, and stress/strain fields. Verification stages (Steps 5–6) validate tables and derivatives against direct pycalphad calculations and compare chemical against elasto-chemical simulations.

Figure 1. Workflow of `PhaseFoundry`. From a CALPHAD TDB (assessed or MLIP-generated via `PhaseForge`), SQS/MLIP elastic data, and a single YAML configuration file, `PhaseFoundry` samples the database with analytic Jansson derivatives, fits a free-energy surrogate, and maps spinodal regions through gauge-invariant Hessian eigenanalysis (pink, `PhaseFoundry.thermodynamics`). The pseudo-spectral Cahn–Hilliard solver (blue, `PhaseFoundry.phasefield`) evolves the instability with an Eyre-stabilized spectral update, optionally coupled to Moulinec–Suquet coherency strain (blue, `PhaseFoundry.elasticity`). Outputs include 2D and 3D spinodal microstructures, composition fields, stress/strain fields, and driving force maps (green). Verification stages (Steps 5–6) validate thermodynamic tables and derivatives against direct `pycalphad` calculations.

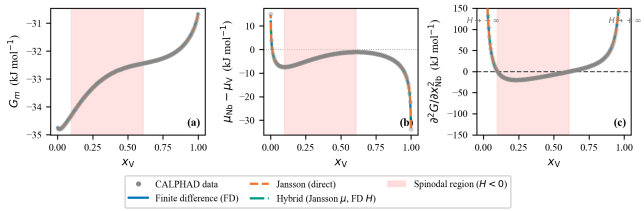


Figure 2. Step 5 validation output for the Nb–V BCC_A2 system at 500 °C: (a) molar Gibbs energy, (b) relative chemical potential $\mu_{\mathrm{Nb}} - \mu_{\mathrm{V}}$, and (c) Gibbs energy Hessian, each computed with finite-difference, direct Jansson, and hybrid derivative modes and overlaid on sampled CALPHAD data. The shaded band marks the spinodal region.

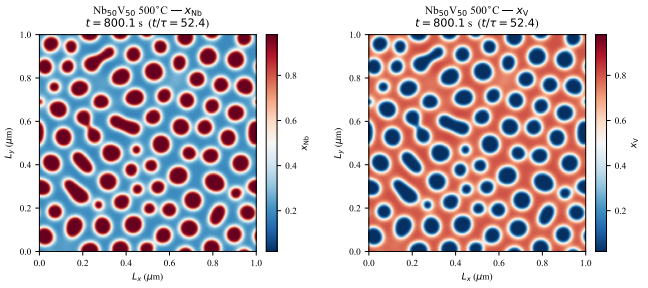


Figure 3. Spinodally decomposed microstructure of equiatomic $Nb_{50}V_{50}$ at 500 °C: Nb (left) and V (right) composition fields on a 1 ţm domain after $t = 800$ s, showing phase compositions consistent with the CALPHAD tie-line.